RES-5 tail-budget closure: the screened higher-skeleton tail fits the

TECT verification-first repository · claim B1-RH-ENUM

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1. Purpose and scope: the budget and the screened tail

The sunset-norm certificate (v1.1) fixed the target: all-orders selection at the operating point survives iff the higher-skeleton ($\ell \geq 3$) pattern-dependent cost fits the third-cumulant slack,

$$C_{\text{higher}} \; < \; \Delta F_{\text{margin}} - C_{\text{leading}} \; = \; \Big(1 - \tfrac{1}{\text{joint}}\Big)\Delta F_{\text{margin}}, \qquad \text{joint} = \frac{\Delta F_{\text{margin}}}{C_{\text{leading}}}.$$

The 2PI screened-response analysis (res5-a0-skeleton-sensitivity; res5-2pi-resummation-strategy) estimates the resummed pattern-dependent tail by the screened response $C_G = 1/(1+g)$ acting on the pattern-dependent dressing amplitude a_0,

$$\boxed{\frac{C_{\text{higher}}}{\Delta F_{\text{margin}}} \; \approx \; C_G \, a_0(I), \qquad a_0(I) = \frac{2\lambda' I}{\hat{r}(I)}, \quad \hat{r}(I) = r_R + 2\lambda' I}$$

with $g = \lambda' B_d = 1.03$ (marginal), $C_G = 1/(1+g) = 0.492$ (repulsive screening, NOT $1/(1-g)$). Since $\hat{r}(I) \to r_R$ for $\lambda' I \ll r_R$, $a_0(I)$ is approximately proportional to I: the tail HALVES from the endpoint to $I = 10^{-3}$.

2. Intensity dependence (the robust finding)

The third-cumulant slack is set by the SC-SCOPE joint, which grows rapidly off the $I = 2 \times 10^{-3}$ endpoint: the endpoint joint is the certified thin 1.040 (scscope-quartic-normalisation-certificate), while the all-orders lift is estimate-feasible with paired joint ~ 3.1 at $I = 10^{-3}$ (scscope-scope-decision) and the STEP-5B floor is $\times 59.4/\times 8.8/\times 2.6$ at $\{4 \times 10^{-4}, 10^{-3}, 2 \times 10^{-3}\}$. Tabulating the budget:

I	$a_0(I)$	tail $= C_G a_0$	joint	slack	tail/slack
4×10^{-4}	0.021	0.010	~ 8	0.875	0.012
1×10^{-3}	0.050	0.025	~ 3.1	0.677	0.036
2×10^{-3}	0.096	0.047	1.040	0.0385	1.224

Off the endpoint the tail shrinks AND the slack grows, so tail/slack improves by $34\times$ from $I = 2 \times 10^{-3}$ to $I = 10^{-3}$.

3. Verdict

Closes for $I \le 10^{-3}$ (STRONG EVIDENCE). The budget holds with a $\ge 27\times$ margin (tail/slack ≤ 0.036); the closure is robust to the estimate uncertainty in both $C_G a_0$ and the off-endpoint joint, because the margin is two orders of magnitude.
Marginal / estimate-undetermined at the $I = 2\times10^{-3}$ endpoint (OPEN). The screened tail 0.047 and the thin slack 0.0385 are the SAME ORDER (tail/slack = 1.22); the all-orders endpoint selection cannot be decided at estimate grade – it could go either way within the uncertainty of $C_G a_0$ and of the certified joint.
RES-5 is localised to the endpoint. It is not a global obstruction: the higher-skeleton tail respects the budget everywhere except the $I = 2 \times 10^{-3}$ endpoint, which is exactly where SC-SCOPE is B1's named second-cumulant hypothesis. This sharpens RES-5/GAP-2 from "OPEN" to "closed for $I \le 10^{-3}$; endpoint-marginal". No tier flip: B1-RH-ENUM stays T6 CONDITIONAL on {H-LAYER}.

4. Estimate vs theorem (CLAUDE.md 6.3.5b)

Both $C_{\text{higher}}/\Delta F_{\text{margin}} \approx C_G a_0$ and the off-endpoint joints are CLASSIFIED ESTIMATES, not constant-bound theorems. Two independent estimates of the endpoint tail coexist in the record and are not yet reconciled: the SC-SCOPE leading cost/margin = 1/joint = 0.962 (slack 0.038) and the resummed $C_G a_0 = 0.047$; they straddle the budget. The $I \le 10^{-3}$ closure does not depend on this reconciliation (its margin absorbs it); the endpoint does. The residual is therefore a single rigorous endpoint bound: $C_{\text{higher}} \le \mathcal{C}\,\|F\|^2$ with $\mathcal{C}$ below the slack, from a genuine 2PI computation of $\Delta\Gamma_2^{\text{pd}}[G_*]$ – or acceptance of the endpoint as the named-hypothesis boundary (the current B1 T6 scope).

5. Devil's-advocate / self-adversarial review (CLAUDE.md 6.3)

(α) *"$C_G a_0$ omits an $O(1)$ skeleton-norm prefactor, so the endpoint tail is under-stated."* VALID: the prefactor (the pattern-dependent self-energy norm relative to the margin) is the residual already flagged in the a0-skeleton note; with it $\gtrsim 1$ the endpoint only worsens, reinforcing the OPEN verdict there. It does NOT affect the $I \le 10^{-3}$ closure, whose $27\times$ margin survives any $O(1)$ prefactor.

(β) *"The off-endpoint joints (~ 3.1, ~ 8) are estimate- grade, so the closure is not rigorous."* VALID-with-mitigation: the closure is labelled STRONG EVIDENCE, not a theorem; the $\ge 27\times$ margin is the reason a factor-few uncertainty in the joint cannot overturn it. A rigorous interval joint(I) on $[4 \times 10^{-4}, 10^{-3}]$ is the registered polish.

(γ) *"Localising RES-5 to the endpoint changes nothing – the production anchor IS the endpoint."* DISMISSED as overstated, ACCEPTED in part: the production anchor is $I = 2 \times 10^{-3}$, so the endpoint marginalit-y is load-bearing for the all-orders claim there – which is precisely why SC-SCOPE is retained as the named hypothesis at the endpoint (B1 T6). The localisation is nonetheless a real advance: it proves the obstruction is one boundary point, not the bulk, and it identifies the off-endpoint regime where Reading-H selection is all-orders strong-evidence.

Quantitative sanity check (mandatory): *scaling* – $a_0(2e\text{-}3)/a_0(1e\text{-}3) = 1.90 \approx 2$ (linear in I); *endpoint* – $0.047/0.0385 = 1.22$ (same order, undetermined); *off-endpoint* – $0.025/0.677 = 0.036$ ($27\times$ margin); *localisation* – $34\times$ improvement endpoint $\to 10^{-3}$.

6. Result footer

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Result ID:                B1-RH-ENUM / RES-5 tail-budget closure
Precise statement:        The higher-skeleton tail budget C_higher < Delta F_margin
                          - C_leading is quantified across the three certified
                          intensities by C_higher/Delta F_margin ~ C_G a0(I),
                          C_G=1/(1+g)=0.492, a0(I)=2 lam' I/rhat ~ I. CLOSES with
                          >=27x margin for I<=1e-3 (tail/slack 0.012, 0.036);
                          MARGINAL/estimate-undetermined at the I=2e-3 endpoint
                          (tail 0.047 vs slack 0.0385, ratio 1.22). RES-5 localised
                          to the endpoint = SC-SCOPE named-hypothesis boundary.
Scope:                    Three certified intensities; estimate-grade (C_G a0 and
                          off-endpoint joints are estimates, not theorems).
Dependencies:             res5-sunset-selfenergy-norm-certificate v1.1 (budget
                          target); res5-a0-skeleton-sensitivity (C_G, a0);
                          scscope-quartic-normalisation-certificate (endpoint joint
                          1.040); scscope-scope-decision (I<=1e-3 feasibility).
Evidence grade:           ANALYTIC (scaling) + EXECUTED (res5_tail_budget.py 5/5)
                          + INHERITED (SC-SCOPE joints, estimate-grade).
Reproduction command:     python codes/vacuum/res5_tail_budget.py
Expected output:          tail/slack = 0.012 (4e-4), 0.036 (1e-3), 1.22 (2e-3);
                          34x localisation; 5/5 PASS.
Falsification gate:       A rigorous endpoint 2PI bound C_higher >= slack (selection
                          overturned all-orders at the endpoint); or an off-endpoint
                          joint << 3.1 that breaks the I<=1e-3 closure.
Tier before / after:      No tier flip. B1 T6 on {H-LAYER}. RES-5/GAP-2 sharpened:
                          OPEN -> closed for I<=1e-3 (strong evidence), endpoint-
                          marginal at I=2e-3.
No-overclaim statement:   This does NOT close RES-5 at the endpoint and does NOT
                          flip B1. It localises the obstruction to the I=2e-3
                          endpoint and establishes strong-evidence all-orders
                          closure for I<=1e-3. Endpoint tail and slack are estimate-
                          grade and straddle the budget (undetermined).
Next required action:     A single rigorous endpoint bound C_higher <= C ||F||^2
                          with C below the slack (genuine 2PI Delta Gamma_2^pd[G_*]
                          computation), OR acceptance of the I=2e-3 endpoint as the
                          SC-SCOPE named-hypothesis boundary (current B1 T6 scope).
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